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A β-glucan-conjugate vaccine and anti-β-glucan antibodies are effective against murine vaginal candidiasis as assessed by a novel *in vivo* imaging technique

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ABSTRACT

The protective capacity of a parenterally administered β-glucan-conjugate vaccine formulated with the human-compatible MF59 adjuvant was assessed in a murine model of vaginal candidiasis. To monitor infection, an *in vivo* imaging technique exploiting genetically engineered, luminescent *Candida albicans* was adopted, and compared with measurements of colony forming units. The vaccine conferred significant protection, and this was associated with production of serum and vaginal anti-β-glucan IgG antibodies. Vaginal IgG molecules were the likely mediators of protection as inferred by the efficacy of passive transfer of immune vaginal fluid and passive protection by an anti-β-1,3-glucan mAb. Overall, the *in vivo* imaging technique was more reliable than vaginal CFU counts in assessing the extent and duration of the vaginal infection, and the consequent protection level.

1. Introduction

Fungal infections have become increasingly common and represent a growing health problem in patients with weakened immune system [1–3]. In addition, some types of mucosal infection undergo multiple recurrences even in non-immunosuppressed subjects, of which a typical example is the recurrent or chronic vulvovaginal candidiasis (RVVC) [4–8]. It has been calculated that around 5% of women suffering an attack of vulvovaginal candidiasis (VVC) will acquire a chronic infection. Although maintenance chemotherapy regimens are useful to counteract any single recurrence, this infection remains substantially incurable [4,9–13].

Candida albicans is the major agent of VVC and RVVC. This fungus is found as relatively harmless commensal in the vagina of most women, and it is not clear which immunological mechanism(s) inhibits the shift from safe commensalism to vaginal infection. Indeed, there is no clear evidence of the type of immunological deficit that predisposes women to acute infection and, in some cases, to recurrent infection. Any deficit must, however, be local since subjects with RVVC do not show appreciable differences with respect to healthy counterparts in systemic immune responses to the fungus [14–16]. The perturbation of indigenous flora, genetic predisposition, imbalances or polymorphisms of some factors of innate immunity, high susceptibility to neutrophil influx are all thought to predispose to RVVC [15,17–20].

In an estrogen-dependent rat model of vaginal candidiasis, vaginal infection has been shown to be associated with hyphal growth and the expression of virulence factors, among which secretory aspartic proteinases play a major role [21–24]. This indirectly supports the reported relationship between severity of clinical symptoms of vaginal infection in women and amount of a secreted aspartic protease in the vaginal fluid [22]. Moreover, it has been found that both in rat and in mouse models, vaginal infection can be controlled by local or systemic vaccination, by immunotherapy with antibodies against virulence factors or adoptive transfer of various kinds of immune cells [25–33]. All of this raises the possibility that VVC and RVVC in women may be controllable by immunopreventive or immunotherapeutic interventions.

To assess the preclinical efficacy of such interventions, a reliable measure of the local infection is an obvious prerequisite. Valuable results have been obtained by counting vaginal colony forming units (CFU) of *C. albicans* both in the vaginal secretion and

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